CV - Dhananjay Tiwari

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Summary

PhD Candidate with 3+ years of experience in optimization, planning and scheduling on large-scale networks, supported by a solid mathematical foundation in analysis, reinforcement learning, probability theory, information theory, and practical experience in building scalable computational frameworks in Python.

Education

University of Illinois at Urbana-Champaign (UIUC) [GPA - 3.9/4.0] **[Aug '22 - May '27]**
PhD in Mechanical Engineering, MS Mathematics (ongoing) | Advisor: Prof. Srinivasa M. Salapaka
- **Best Presenter Award**, CSL Student Conference 2025, MDPs, RL and Optimization Session

Indian Institute of Technology Bombay (IIT Bombay) [GPA - 9.13/10] **[Jul'17 - Jun'22]**
B.Tech & M.Tech in Mechanical Engineering | Advisor: Prof. Ravi N. Banavar
- **Institute Academic Prize** (2018-2019) for the best peformance in a batch of 40 in the academic year
- **Bronze Medal** among 20+ participants at Inter-IIT Tech Meet 2018
- Minor degree in Systems and Control Engineering

Publications

1. Basiri S, **Tiwari D**, Hassani G, Papachristos C, Salapaka S. A Maximum Entropy Approach to Joint Obstacle-Aware Routing and Facility Location in UAV Networks Under Energy Constraints. IEEE Access. 2025 Dec 24;13:216694-704.
2. **Tiwari D**, Basiri S, Salapaka S. Multi-Agent Integrated Path Planning and Scheduling in Advanced Air Mobility. InAIAA SCITECH 2025 Forum 2025 (p. 1352). [DOI: 10.2514/6.2025-1352]
3. **Tiwari D**, Banavar RN. Discrete geometric control of planar flexible link manipulators. IFAC-PapersOnLine. 2023 Jan 1;56(2):2865-70. [DOI: 10.1016/j.ifacol.2023.10.1402]
4. Bayati A, **Tiwari D**, Salapaka S. A Control Barrier Function Approach to Constrained Resource Allocation Problems in a Maximum Entropy Principle Framework. arXiv preprint. 2025 Apr 2. [DOI: 10.48550/arXiv.2504.01378]
5. Basiri S, **Tiwari D**, Papachristos C, Salapaka S. Optimizing UAV Network Efficiency: Integrative Strategies for Simultaneous Energy Management & Obstacle-Aware Routing. AIAA SCITECH 2024 Forum 2024 (p. 1166). [DOI: 10.2514/6.2024-1166]
6. **Tiwari D**, Basiri S, Salapaka S. Towards Enabling Learning for Time-Varying finite horizon Sequential Decision-Making Problems. arXiv preprint arXiv:2504.02129. 2025 Apr 2.
7. Herrmann M, **Tiwari D**, Kotyczka P, Banavar R. Coordinate-Invariant Modeling and Control of a Three-DOF Robot Manipulator. IFAC-PapersOnLine. 2021 Jan 1;54(19):230-6. doi: [DOI: 10.1016/j.ifacol.2021.11.083]
8. **Tiwari D**, Banavar RN, Maithripala DH. *Geometric Modelling and Energy Optimisation in a Heaving-Point Absorber Wave Energy Converter*. In2021 29th Mediterranean Conference on Control and Automation (MED) 2021 Jun 22 (pp. 967-972). IEEE. [DOI: MED51440.2021.9480198]

Research Experience

Optimization and Planning | Advisor: Prof. Srinivasa M. Salapaka (UIUC), Prof. Ankur Mani (UIUC)

Distributed Online Stochastic Linear Optimization [Under Review: ACM EC 2026] **[Oct '25 - Feb '26]**
- Formulated an online stochastic linear optimization framework with **bandit feedback** involving myopic agents guided by a central planner.
- Designed a novel **subspace contraction policy** enabling the planner to steer agents toward the optimal decision.
- Proved problem-independent **regret bounds** of $\mathbf{O}(\mathbf{d}^{3.5}\log\mathbf{T})$ for a general polytope as a decision space.

Simultaneous Multi-Agent Routing and Scheduling [AIAA Scitech 2025] **[Jan '24 - Present]**

- Built a **modular Python framework** for conflict-free multi-agent routing and scheduling on graph networks, optimizing **makespan** and **total travel cost** across large networks.
- Achieved **10× faster runtime** compared to penalty based baseline by designing an **annealing-inspired optimization algorithm** integrated with control barrier functions.

Parametrized Multi-Agent Routing via Deep Attention Models [ArXiv] **[Mar '25 - Aug '25]**

- Designed a hybrid shortest-path and uniform-path sampling scheme for multi-agent routing and facility location, achieving **scalability of up to 10×** compared to baselines at comparable cost and faster runtimes.
- Accelerated gradient based optimization by $\mathbf{O(M^2)}$, with **10× lower costs** compared to existing metaheuristics; achieved **1500× speedup** compared to Gurobi's exact mixed-integer programming solver.

Capicitated Resource Allocation [CDC 2025] **[Nov '24 - Mar '25]**

- Designed a **nonlinear optimization framework** via **Maximum Entropy Principle** and **Control Barrier Functions**, guaranteeing **KKT** conditions while efficiently searching in a large combinatorial space.
- Achieved **71× and 4× speedup** compared to SciPy's SLSQP & Safe Gradient Flow with efficient constraint handling.

Dynamics and Controls | **Advisors: Prof. Ravi N. Banavar (IIT Bombay), Dr. Paul Kotyczka (TU Münich)**

Geometric Modelling & Control of Flexible Link Manipulators [IFAC WC 2021] **[Jul '21 - Jul '22]**

- Designed a control law for point-to-point manoeuvre of flexible manipulator with suppressed flexural deflections.
- Developed a **Lie group variational integrator** model for a flexible manipulator.
- Performed a preliminary convergence analysis of LGVI for the evolution of beam cross-section rotations

Coordinate Invariant Modelling & Control of a 3-DOF manipulator [LHMNC 2021] **[Jul '20 - Jun '21]**

- Developed a discrete-time coordinate-free modeling and control algorithm for a 3-DOF, revolute-joint manipulator.
- Simulated the closed loop discrete model using geometric **potential shaping** algorithm for joint angle control.
- Validated structure-preserving property of the integrator through simulations; obtained **average deviation in the order of** 10^{-14} for all the configuration variables with reference to the robot's configuration space.

Energy Optimisation in Wave Energy Converters Devices [MED 2021] **[May '19 - Mar '20]**

- Conceptualized WEC as floating sphere with internal 3D-pendulum to harvest useful energy from ocean waves
- Derived a dynamic model for 3D motion of a **double pendulum** and validated it through numerical simulations.
- Formulated and solved an **optimal control** problem on CasADi-MATLAB framework, aimed to maximize the energy conversion due to relative motion between spherical shell and pendulum with limited actuation capacity.

Teaching Experience

Instrumentation and Signal Processing (UIUC) **[Jan '23 - Present]**

- Led **100+** students in lab experiments worth of **130+** hours on signal processing and real-time control
- Designed homework and supported students through office hours; taught theory classes.

Microprocessors and Automatic Control (IIT Bombay) **[Aug '21 - Nov '21]**

- Assisted **15+** students to operate lab microcontrollers and peripheral setup through a virtual environment; linked theoretical concepts to practical implementations.

Engineering Mechanics (IIT Bombay) **[Jan '20 - Mar '20]**

- Mentored **70+** students through tutorials, improving conceptual understanding and problem-solving skills.

Mentorship and Leadership Experience

Student Mentor, Student Mentorship Program, **[Oct '20 - Jul '21]**

- Mentored **14** first year undergraduates to ensure their smooth intellectual and professional development
- Received **ISMP Special Recognition award**, highest honour bestowed internally by the institute body

Controls Subsystem Lead, Student Satellite Program, IIT Bombay **[May '19-May '21]**

- **Led** a team of **8** to design a simulation framework for **autonomous rendezvous & docking** of two spacecrafts. **Recruited** the members by designing technical assessments, conducting interviews, and guiding them through two-week project-based onboarding.
- Created a **Python-based simulation framework** for the student satellite community, aimed to provide a testing platform for Attitude Determination and Control algorithms in a simulated space environment

Manager, Controls & Dynamical Systems Student Reading Group, **IIT Bombay** **[Apr '20 - Apr '21]**

- Conceptualised & organised a series of lectures to introduce newcomers to key concepts in control engineering
- Ideated and coordinated the **Summer Learning Projects-2020**, which provided a platform for over **20+ students** to work collaboratively on fundamental topics from various domains of control theory

Selected Talks and Public Engagements

Paper Reviewer	AAAI 2026, IEEE L-CSS	**[Aug '25-Oct '25]**
Voluntary High-School Research Mentorship	UIUC, Champaign, IL	**[Jun '25-Sep '25]**
Invited Speaker - Routing and Scheduling	Tata Consulting Services Research, Mumbai	**[Jan '25]**
Invited Speaker - Intermediate Axis Theorem	Maths & Physics Club, IIT Bombay	**[Jun '21]**
Speaker - Lectures in Differential Geometry	System & Control Engineering, IIT Bombay	**[Aug '21]**
Geometric Mechanics Workshop	Systems and Control Engineering, IIT Bombay	**[Aug '19]**
Nonlinear Control & Robotics Workshop	University of Peradeniya, Sri Lanka	**[Mar '19]**
Helicopters Lab Visit	Aerospace Engineering, IIT Kanpur	**[May '19]**

Technical Skills

Programming	Embedded C/C++, Python, PyTorch & NumPy (Vectorization & Broadcasting), CVXPY, MATLAB/Simulink, Git/GitHub, CAD.
Electronics	TI Microcontrollers, NI USB-7855R, Optical Encoders, MPUs, LabVIEW FPGA.
Hardware	Atomic Force Microscopy, Self-Balancing Segway Bot, Automated Strawberry Picker Bot
Analytical	Model Predictive Control, Adaptive Control, Convex Optimization, Dynamic Programming, Reinforcement Learning, Kinematic Modeling and Simulation.

Extracurricular

Fine Arts **10+ years** of self-taught experience in realistic portrait and architectural drawing with graphite, ink, and charcoal. **Winner of Kalamanch** (Fine Arts, Abhyuday–IIT Bombay). **Featured for 3 years** in Kaladarshan, the Fine Arts Club's flagship exhibition at IIT Bombay.